

Multi-narrative Serious Game Design Applying Learning Style Model for Cultural Heritage Education

Conceptual Frameworks: The Guardians 2.0

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Q. How to give people meaningful, valuable and exciting learning experiences about cultural heritage?





Difficulties in Cultural Heritage Education

Digital generations think it is a tedious and difficult

→ Games can be a platform of learning for digital generations

Learning by doing without time and space constraints

The Guardians : Serious Game Design for Enhancing Awareness of Cultural Heritage Preservation

Jimin Rhim, et.al (2013).

The Guardians: serious game designed for learning cultural heritage and promoting the awareness of its preservation





In ver.1, same linear narrative experience were given to all players,
no matter what character they chose

But People have…

different preference for learning activities
different point of view when learn something



Customized learning experience → Increased interest and learning effect

The Guardians 2.0 ?

4 characters → 4 learning styles!





Take the object as a UNESCO world heritage in KOREA

"Royal Tombs of Joseon Dynasty"

“Royal Tombs of Joseon Dynasty”

Root of current Korea

Special stories of the kings

Values and ideologies of the time

Documentations about information including construction guideline and memorial service processes

→Identity of Korea!

Many are located in near or in the cities,
but many Korean people cannot get the meaning of them.



A scene of one of the Royal Tombs
(a red gate with spiked top and
a T-shaped housefor sacrifice)



A gameplay screen of The Guardians 2.0

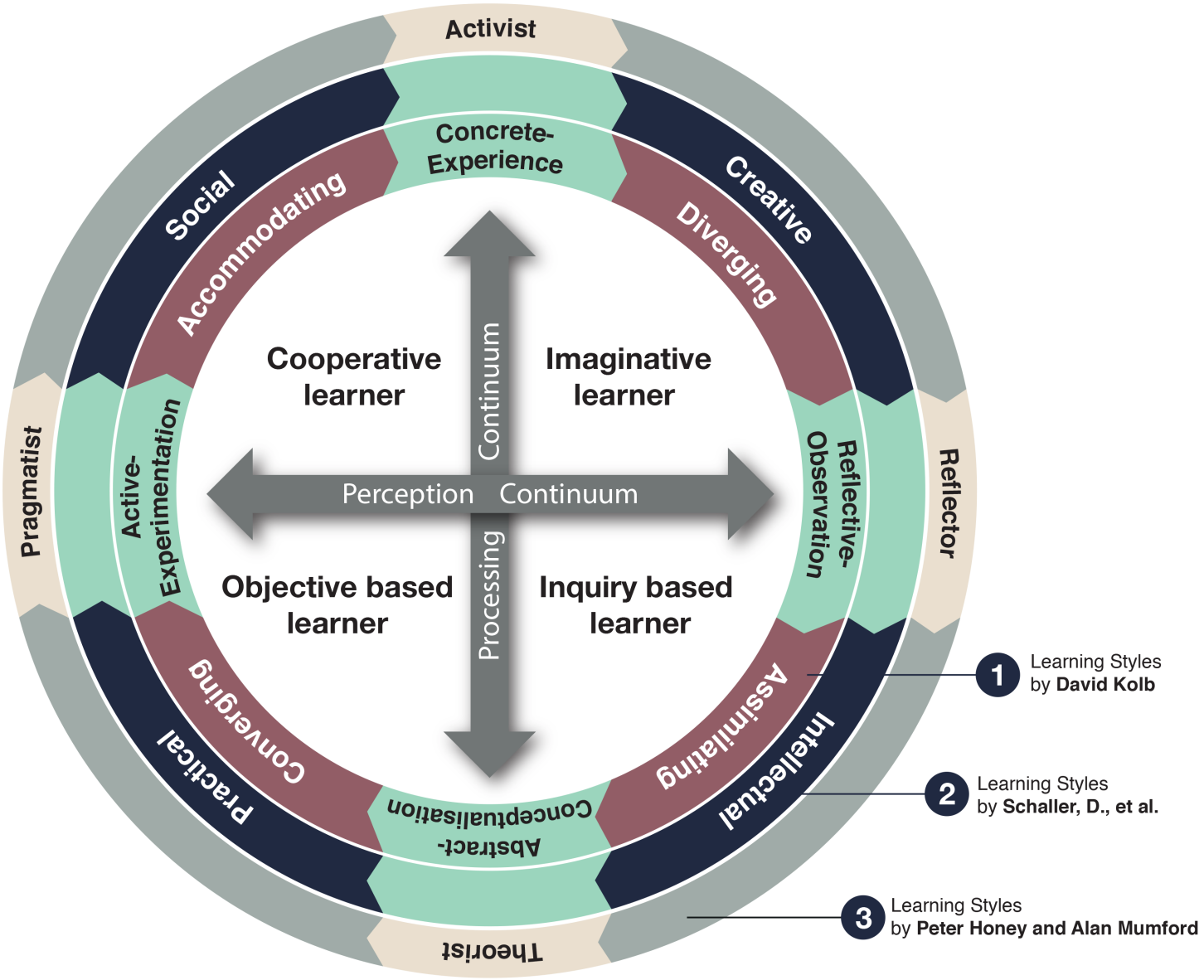
Integrated learning style model for The Guardians 2.0

Try to give people activities suit for their characteristics

Match three learning style theories

- Kolb's Experiential learning theory
- Schaller's learning style model for museum visitors
- Learning style model of Honey and Mumford

Put our preferred label for four learning styles



Narrative in Serious Games

Way of achieving an entertainment aspect
Delivering message
→ tried to reinforce the narrative for enhancing the learning effect

set the first-person view:

Players identification with the main character

4 characters:

Pedagogical agents have different learning style, and different narrative paths
→ multi-narrative structure



game narrative frame have character-specific playing routes

Learning Style	Imaginative Learner	Inquiry based Learner	Objective based Learner	Cooperative Learner
Character	Photographer	Historian	Treasure Hunter	Traveller
Learning Activities	Careful observation & unrestricted exploration	Finding question & answer following logical way	Active experiment & Abstract conceptualization	Learn by talking and d iscussion & seek new experience
Layer of Cultural Heritage Information	aesthetic view	historical background	traditional ideology	rituals in present





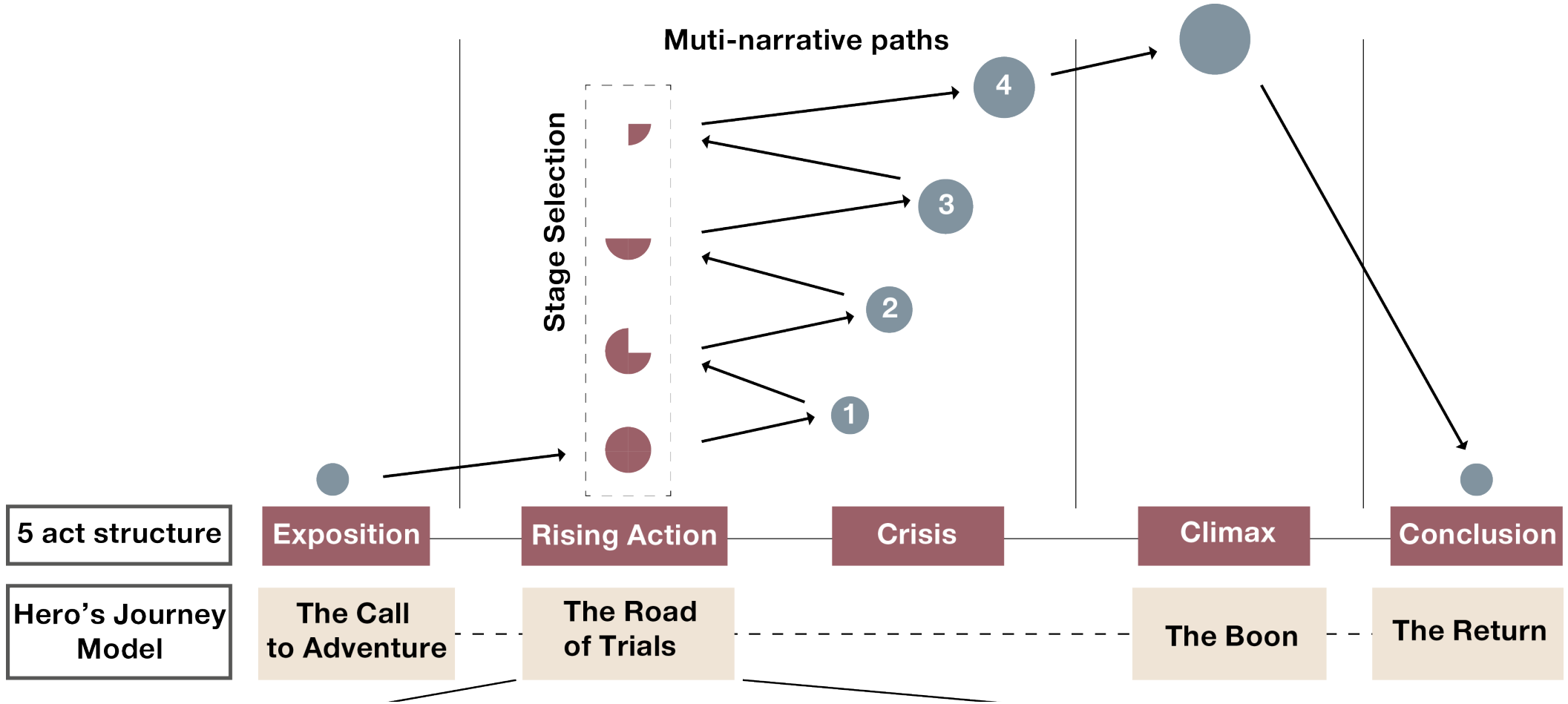




Multi-narrative structure frame for Adventure Genre Serious Games

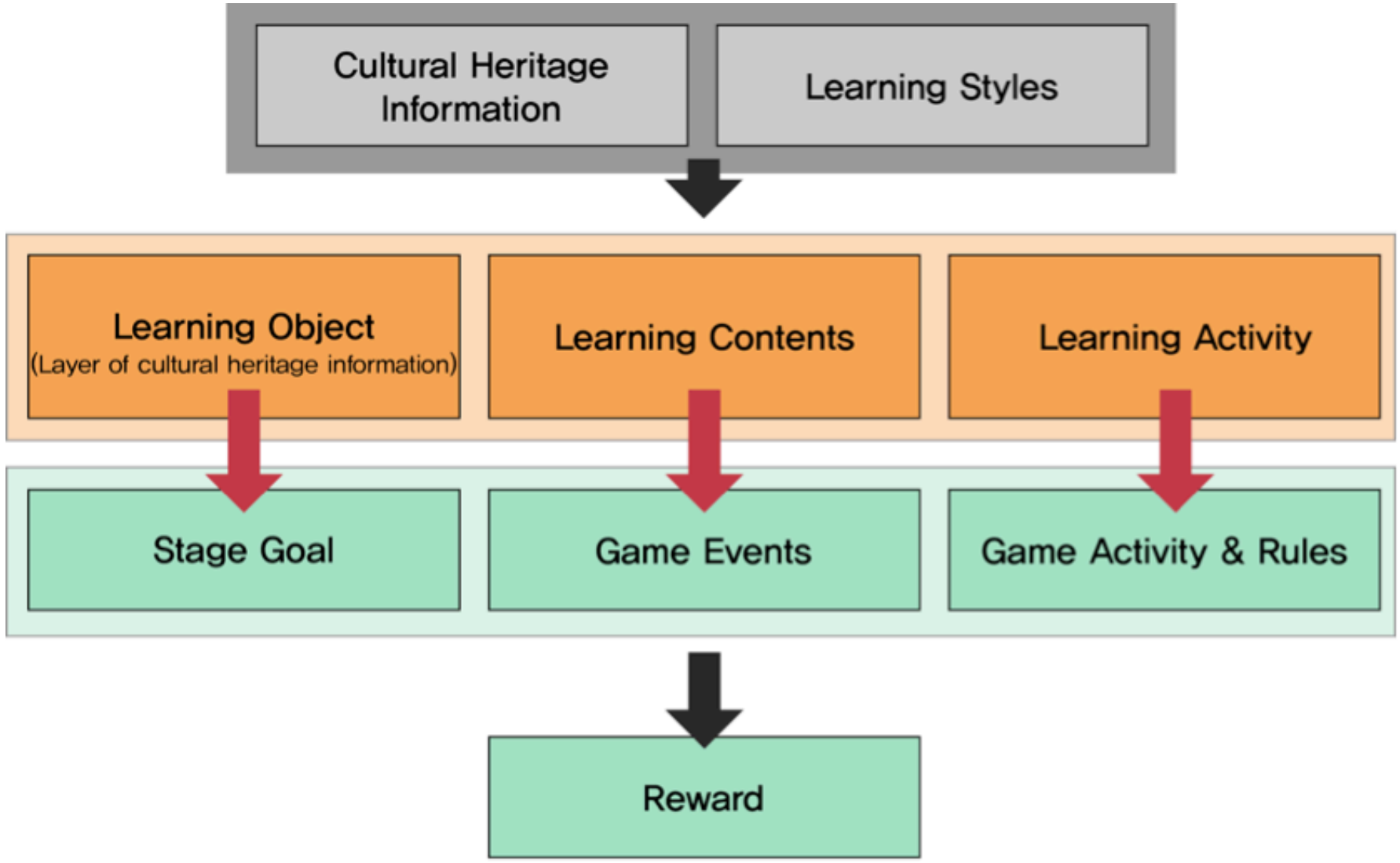
Graft ‘hero’s journey’ model onto a five-act structure

Rising action(the road of trial): match learning elements with game elements for details



A Framework for designing each narrative path and game mechanics components in The Guardians 2.0

Layer of cultural heritage information
- aesthetic view
- historical background
- traditional ideology
- rituals in present



Cultural Heritage Information

The formative information of cultural heritages.
Visual information and formative meanings.

Learning Style

Imaginative learner

Learning Object

(Layer of cultural heritage information)

Learn about meaning of building and design of Royal tombs of Joseon dynasty
Learning contents

Learning Contents

(1) The function and role of a monument house of Joseon Royal tombs
(2) The formative meanings of stone figures of Joseon Royal tombs

Learning Activity

Careful observation
& Unrestricted Exploration

Stage Goal

Assist photographer guardians and restore the stone figures. Then taking a picture of Royal tombs of Joseon dynasty!

Game Events

(1) You have to find and take a picture of monument house. But there is a disrupt element(stain) in monument.

(2) You have to take a picture of stone figures. Some stone figures

Game Activity & Rules

(1)
– Click / Cleaning the monument house of Joseon Royal tombs.
– Click / Taking a picture
(2)
– Click / a picture puzzle
– Click / Rotate and shift the direc–



Narrative frame example of Photographer's route

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Visual information and formative meanings.

Imaginative learner

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(2) You have to take a picture of stone figures. Some stone figures are in wrong direction. Find them and return before taking a picture.

Game Activity & Rules

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– Click / Rotate and shift the direc–tion of the stone figures. Then take a picture.
– Click / Taking a picture



Narrative frame example of Photographer's route

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Reward

Real picture of Joseon Royal tombs. User can share the pictures in SNS.

Mission Narrative

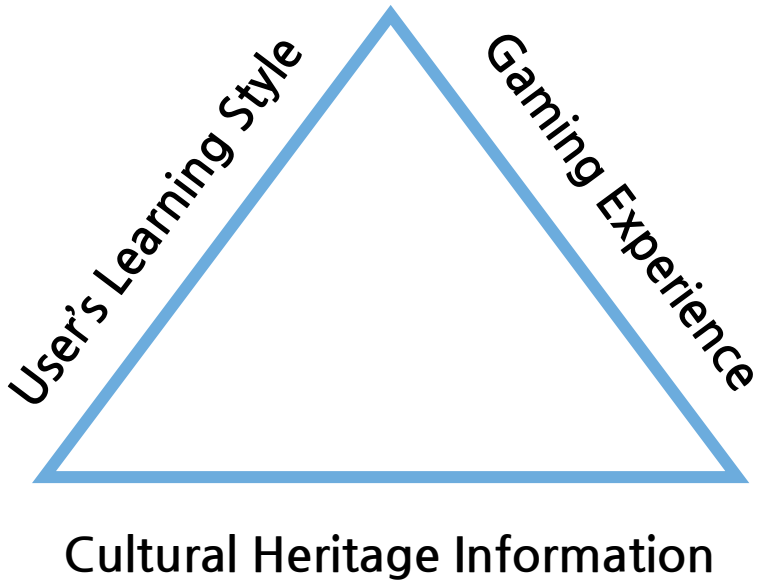
Someone spoil image database and stone figures of Royal tombs of Joseon dynasty. That's the reason of decreas–ing spirit energy of Royal tombs! Find the damaged stone figures and restore. Assisting pho–tographer guardians, take a picture of restored Royal tombs of Joseon dynasty!



Narrative frame example of Photographer's route

Implication

- For interesting learning experience to understand cultural heritage, we make a multi-narrative frame for a serious game 'The Guardians 2.0'



- The conceptual frame structures can be applied in any field

Future Work

Improve the frameworks

Detail the frames for constructing narrative and game activity

Implement a prototype and conduct user study

does it work?

The ultimate aim of “The Guardians” series :

Make an interesting and meaningful game
that make players participating in cultural heritage preservation
both in game and real world

Thank You

Questions and Comments, Please

A Concept work of a Prototype: The Guardians 2.0

- Genre : 2D first person view adventure game
  - First person view: Player's identification with a main character
  - Adventure games: Engaging storytelling, challenging puzzles.

- Main Story
  - Muinseok, a statue that protects the Royal Tombs of the Joseon Dynasty, ask p layers for help with a threatening for the Royal Tombs.
  - To save the royal tombs, players visit The Guardians office and get missions to repair the damaged tombs with four guardians.
  - In the recovery process of the spiritual energy of the royal tombs, the players a re faced with a villain who damages the tombs.
  - Finally, the guardians and players defeat the villain and succeed in protecting t he royal tombs, and each player gets certificate of the guardians.